building blocks, we will be able to create an advanced data structure that can be used to quickly access values/computed results that were recently accessed – although one could argue that a cache is more than just a data structure, it's already a sophisticated component with several moving parts: in its simplest form, it can just be implemented as an associative array, but we will see during the course of this chapter that it can become as complex as a web service.

In this chapter we are going to delve into the details of how caches of increasing complexity work, so that, after reading it, you should be able to articulate an informed opinion on whether the topic.

7.1 Don't Compute Things Twice

In our daily job as software developers, we daily write applications, most of which perform really simple tasks most of the time. Adding two numbers, or even dividing them or adding two vectors (with modern GPU[1]s) are trivial operations, fast enough that we don't need to bother about optimizing them (of course this wasn't always the case: but if you happen to be young enough, you never had to deal with x86 processors...).

Yet, no matter how faster and optimized multicore processors or server clusters become, there will always be some kind of computations, some complex operations that will just be too